Applications of fractions

1 Sharon walks $\frac{2}{5}$ of a kilometre and runs $\frac{1}{7}$ of a kilometre to get to school. How many kilometres does she travel?

2 Carl has $\frac{3}{7}$ m of ribbon. After he uses some ribbon for a present, he has $\frac{1}{4}$ m left.

How much ribbon did he use on the present?

3 For a project, Ivan needs to cut a metal rod such that the longer piece is three times the length of the shorter piece.

a What fraction of the rod's total length will be made up of the shorter piece?

b If the rod is $\frac{3}{8}$ m long, what is the length of the shorter piece?

4 Bianca has $\frac{5}{6}$ kg of beads. She shares the beads evenly between 3 friends. How many kilograms does each friend receive?

5 At a party, Bill makes a drink by combining $5\frac{1}{3}$ L of water with $1\frac{1}{2}$ L of cordial. What is the total amount of litres in the mixed drink?

6 Skye wants to decorate her window sill with fairy lights. The window sill is $3\frac{4}{9}$ m wide. If she only has $2\frac{1}{8}$ m of fairy lights, how much of the window sill will not be decorated?

7 Katrina takes $3\frac{2}{3}$ minutes to drive from her home to the local shopping centre. She spends $\frac{1}{4}$ of this time waiting at traffic lights. Find the number of minutes she spends waiting.

8 Jack is making bags for his friends. He has $3\frac{1}{2}$ m of fabric. If each bag requires $\frac{2}{5}$ m of fabric, how many bags can he make?

9 At the Easter Show, Tobias spends $\frac{2}{4}$ of an hour visiting chickens and $\frac{3}{9}$ of an hour visiting alpacas. How much time has he spent visiting animals?

10 Caitlin runs a cafe and is interested in the drink choices of her customers. She finds that 3 out of 4 customers order hot chocolate. If she has 4000 customers on a particular day, how many hot chocolates would she expect to sell?

11 A bag of oranges weighing $\frac{8}{9}$ kg was divided among 6 children. What was the mass of the

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oranges given to each child?

12 Tricia attends a 3-hour painting class. The instructor uses $\frac{12}{14}$ of an hour to discuss colour mixing and then allows students to use the rest of class time to complete a painting. How much time does Tricia have left to finish her work?

13 Katrina and Luigi ordered a pizza to share. Katrina ate $\frac{3}{7}$ of the pizza while Luigi ate $\frac{1}{2}$.

a Who ate more pizza?

b What fraction of the pizza did they eat together?

c What fraction of pizza is left?

14 Luke has a dice tin that is $\frac{1}{4}$ full. He puts 8 extra dice in and the tin is now $\frac{1}{3}$ full.

a What fraction of the tin do the 8 extra dice take up?

b How many dice can fit inside the tin?

15 Victoria borrows a 360-page novel from the library. She reads $\frac{3}{4}$ of the novel.

a How many pages did she read?

b How many more pages does she have to read to finish her novel?

16 Christa is creating a 3 minute slide show with pictures from her holiday. Each picture will be displayed for $4\frac{1}{2}$ seconds.

a How many seconds are in 3 minutes?

b How many pictures can she show in 3 minutes?

17 Danielle is studying for three tests. She studies science for $3\frac{1}{4}$ hours, English for $\frac{2}{5}$ of an hour and visual arts for $\frac{1}{4}$ of an hour. How many hours does she spend studying?

18 Homer works as an accountant and is at the office for $7\frac{1}{2}$ hours each day. He eats lunch with his colleagues for $\frac{5}{6}$ hours each day. How much time does he spend working in one day?

19 In a school debate, each speaker was allowed $1\frac{6}{7}$ minutes. How many speakers participated if the debate went for 26 minutes?

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20 Paul delivers online orders to customers for $6\frac{5}{8}$ hours each weekday.

- **a** Over the five weekdays, how many hours does he work?
- **b** On Saturdays he works half the time of a weekday. How many hours does he work on a Saturday?
- **c** How many hours in total does he work in one week if he does not work on Sundays?

21 Han spent $\frac{1}{8}$ of his money on a printer and $\frac{1}{5}$ of his money on a table.

- **a** What fraction of his total money did Han spend?
- **b** What fraction of Han's money does he still have?

22 In a survey of 270 people, $\frac{1}{10}$ said their favourite sport was soccer, and $\frac{1}{9}$ said their favourite sport was tennis.

- **a** Calculate the total fraction of people who said their favourite sport was soccer or tennis.
- **b** Calculate the fraction of people who did not put soccer or tennis as their favourite sport.
- **c** Calculate the number of people who did not put soccer or tennis as their favourite sport.

23 In a class with 40 students, $\frac{2}{5}$ of the class are boys.

- **a** What fraction of the class are girls?
- **b** How many girls are in the class?

24 An engineer is estimating the length and width of a room using the length of his steps. The length of the room is $\frac{4}{5}$ of the width. He determined that the width is about 27 steps and that one of his steps is $\frac{2}{3}\ \text{m}$ long. Find the length of the room.

25 Two cars leave home at the same time travelling south. Car A travels at the rate of $83\frac{1}{4}\ \text{km/h}$ while Car B is at $88\frac{3}{4}\ \text{km/h}$. If both cars continue to travel at these speeds in the same direction, how far apart will the two cars be at the end of 4 hours?

26 About $\frac{3}{10}$ of the cars in a city are less than 5 years old. About $\frac{1}{5}$ of the cars are between 5 and 10 years old. If there is a total of 1200 cars registered in the city, find the number of cars that are greater than 10 years old.

27 The mall parking lot has 800 parking spaces. $\frac{1}{4}$ of the parking spaces are for cars. On a particular day there were 200 cars and some trucks in the parking lot. The parking lot was $\frac{5}{8}$

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